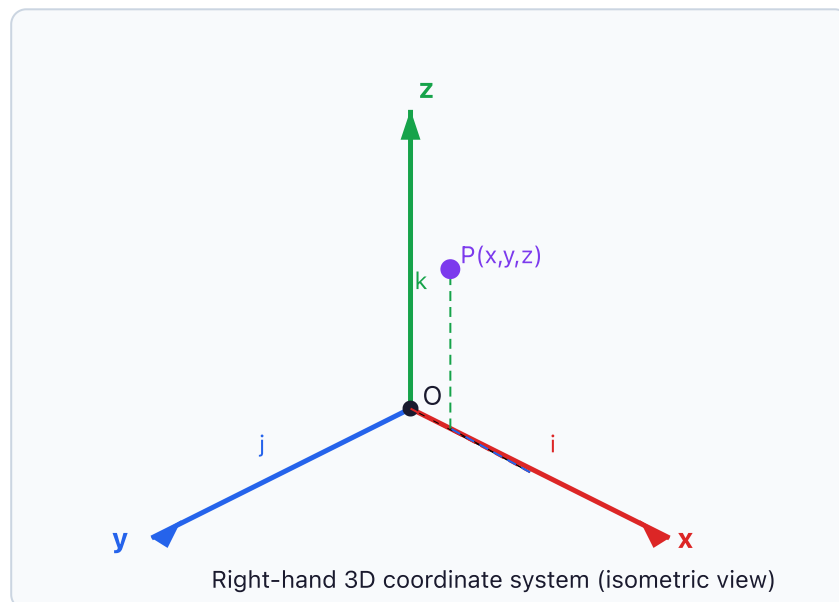


# P21: Vectors (Year 2 — 3D)

## 1. 3D Coordinate System

In three dimensions, every point is described by three coordinates  $(x, y, z)$ . The three coordinate axes ( $x$ ,  $y$ ,  $z$ ) are mutually perpendicular.



## 2. 3D Vectors — Notation and Basics

A 3D vector can be written in two equivalent ways:

**Column vector:**  $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$

**ijk notation:**  $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

where  $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ ,  $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ ,  $\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$  are unit vectors along the coordinate axes.

## Operations on 3D vectors

| Operation             | Formula                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------|
| Addition              | $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$  |
| Scalar multiplication | $\lambda \mathbf{a} = \begin{pmatrix} \lambda a_1 \\ \lambda a_2 \\ \lambda a_3 \end{pmatrix}$ |
| Magnitude             | $ \mathbf{a}  = \sqrt{a_1^2 + a_2^2 + a_3^2}$                                                  |
| Unit vector           | $\hat{\mathbf{a}} = \frac{\mathbf{a}}{ \mathbf{a} }$                                           |

## 3. Position Vectors and Distance

The position vector of point  $P(x, y, z)$  is  $\overrightarrow{OP} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ .

**Vector between two points:**  $\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \\ b_3 - a_3 \end{pmatrix}$

**Distance between A and B:**  $|AB| = |\overrightarrow{AB}| = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}$

**Example:** Find the distance between  $A(1, 2, -1)$  and  $B(4, 0, 3)$ .

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix}, |AB| = \sqrt{9 + 4 + 16} = \sqrt{29}$$

## 4. Equation of a Line in 3D

**Vector equation of a line** passing through point  $A$  with position vector  $\mathbf{a}$ , in direction  $\mathbf{b}$ :

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}, \quad t \in \mathbb{R}$$

The parameter  $t$  gives different points on the line.

### Cartesian form of a line

If  $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ , the Cartesian form is:

$$\frac{x - a_1}{b_1} = \frac{y - a_2}{b_2} = \frac{z - a_3}{b_3} = t$$

**Example:** Write the vector equation of the line through  $A(1, -2, 3)$  with direction  $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ .

$$\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Cartesian form:  $\frac{x - 1}{2} = \frac{y + 2}{1} = \frac{z - 3}{-1}$

**Example:** Find the vector equation of the line through  $P(2, 1, 0)$  and  $Q(5, 3, -2)$ .

$$\text{Direction: } \overrightarrow{PQ} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

**Exam tip:** Any scalar multiple of the direction vector gives the same line. The point on the line can be any point — not just the specific one given. Use the simplest values.

## 5. Scalar (Dot) Product

$$\text{For } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}:$$

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}| \cos \theta$$

where  $\theta$  is the angle between  $\mathbf{a}$  and  $\mathbf{b}$ .

### Finding the angle between two vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

**Example:** Find the angle between  $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ .

$$\mathbf{a} \cdot \mathbf{b} = 0 + 2 - 2 = 0$$

Since the dot product is 0, the vectors are **perpendicular**.

**Example:** Find the acute angle between  $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ .

$$\mathbf{a} \cdot \mathbf{b} = 2 - 1 + 0 = 1$$

$$|\mathbf{a}| = \sqrt{5}, |\mathbf{b}| = \sqrt{3}$$

$$\cos \theta = \frac{1}{\sqrt{15}}, \text{ so } \theta = \cos^{-1}\left(\frac{1}{\sqrt{15}}\right) \approx 75.0^\circ$$

## Perpendicular vectors

$$\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$$

## 6. Angle Between Two Lines

The angle between two lines with direction vectors  $\mathbf{b}_1$  and  $\mathbf{b}_2$  is found using:

$$\cos \theta = \frac{|\mathbf{b}_1 \cdot \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|}$$

**Take the modulus** so that  $\theta$  is the acute angle between the lines (lines are undirected).

**Exam tip:** Always use  $|\mathbf{b}_1 \cdot \mathbf{b}_2|$  (absolute value of the dot product) when finding the angle between two lines. If you get  $\cos \theta < 0$ , the answer would be the obtuse angle — take  $180^\circ - \theta$  instead.

## 7. Parallel, Intersecting, and Skew Lines

| Relationship | Condition | Test |
|--------------|-----------|------|
|--------------|-----------|------|

|              |                                                                               |                                                                             |
|--------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Parallel     | Direction vectors are scalar multiples: $\mathbf{b}_1 = \lambda \mathbf{b}_2$ | Check $\mathbf{b}_1 \times \mathbf{b}_2 = \mathbf{0}$ (or ratio test)       |
| Intersecting | Lines meet at a point                                                         | Set $\mathbf{r}_1 = \mathbf{r}_2$ , solve for parameters; check consistency |
| Skew         | Not parallel, not intersecting                                                | As above — if equations are inconsistent and not parallel, lines are skew   |

## Testing for intersection

1. Write both lines parametrically:  $\mathbf{r}_1 = \mathbf{a}_1 + s\mathbf{b}_1$  and  $\mathbf{r}_2 = \mathbf{a}_2 + t\mathbf{b}_2$
2. Set equal component by component — gives 3 equations in 2 unknowns ( $s$  and  $t$ )
3. Solve two of the equations for  $s$  and  $t$
4. Substitute into the third equation — if consistent, lines intersect; if not, lines are skew

**Example:** Lines  $\ell_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$  and  $\ell_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ . Do they intersect?

Setting equal:

$$x : 1 + s = 2 + 2t \Rightarrow s - 2t = 1 \dots (1)$$

$$y : 2 + s = 1 - t \Rightarrow s + t = -1 \dots (2)$$

$$(1)-(2): -3t = 2 \Rightarrow t = -\frac{2}{3}$$

$$\text{From (2): } s = -1 - t = -1 + \frac{2}{3} = -\frac{1}{3}$$

$$\text{Check } z : 3 + s(-1) = 1 + t(1): 3 + \frac{1}{3} = 1 - \frac{2}{3}: \frac{10}{3} \neq \frac{1}{3}.$$

Inconsistent — the lines are **skew**.

## 8. Key Properties of the Dot Product

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$  (commutative)
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$  (distributive)
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$
- $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1, \mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{i} \cdot \mathbf{k} = 0$

## 9. Summary of Key Formulae

| Topic                 | Formula                                                                      |
|-----------------------|------------------------------------------------------------------------------|
| Magnitude             | $ \mathbf{a}  = \sqrt{a_1^2 + a_2^2 + a_3^2}$                                |
| Unit vector           | $\hat{\mathbf{a}} = \mathbf{a}/ \mathbf{a} $                                 |
| Distance AB           | $\sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}$                       |
| Line equation         | $\mathbf{r} = \mathbf{a} + t\mathbf{b}$                                      |
| Dot product           | $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$                     |
| Angle between vectors | $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a}  \mathbf{b} }$ |
| Perpendicularity      | $\mathbf{a} \cdot \mathbf{b} = 0$                                            |